OmniCAD Verification & Testing Suite

Structured Prompts & 2-Step Conversational CAD Commands for Autodesk Inventor Sync

Target Web Workbench: http://192.168.11.86:9999
CAD Workstation (Sync Target): 192.168.11.150 (Autodesk Inventor 2025)
Local Save Directory: C:\Users\koustubh\Documents\OmniCAD\

Evaluation Objectives:
- Sub-second (<350ms) initial B-Rep generation
- Real-time multi-step parametric editing
- Live 3D assembly / Part auto-load into Inventor

1. Structural & Mounting Plates (Sheet Metal / Machined)

Part: Heavy-Duty Conveyor Bracket | *Validates 4-corner hole coordinate array, dual center slots, and edge corner fillets.*

STEP 1 Creation	`Create a mounting plate 180mm long, 100mm wide, and 12mm thick with 4 M10 corner holes inset by 15mm from each edge`
STEP 2 Edit	`Add two 40mm long adjustment slots along the center and apply 5mm fillets to all 4 vertical corners`
Expected	Rectangular solid plate with 4 counterbore/through holes, 2 stadium slots, smooth rounded corners in Inventor.

Part: Precision Sensor L-Bracket | *Validates multi-face geometry attachment, vertical vs horizontal sketch planes, through-wall cuts.*

STEP 1 Creation	`Create an L-bracket with a base 120mm by 60mm and 10mm thick, and an upright wall 80mm high and 10mm thick`
STEP 2 Edit	`Add two 8mm mounting holes on the base plate and a 25mm diameter center bore on the upright wall`
Expected	90-degree bracket with 2 base mounting holes and a circular cable/sensor cut on the vertical upright.

2. Rotating Machinery & Power Transmission

Part: Stepped Motor Drive Shaft with Keyway | *Validates multi-stage concentric cylindrical union, axial through-bores, and blind keyway slots.*

STEP 1 Creation	`Create a stepped shaft with a large section 40mm diameter and 60mm long, and a smaller section 25mm diameter and 50mm long`
STEP 2 Edit	`Add a 6mm wide, 4mm deep, and 30mm long keyway along the 25mm shaft section and a 10mm center through-bore`
Expected	Dual-diameter drive arbor with concentric through-hole and standard DIN 6885 keyseat.

2. Rotating Machinery & Power Transmission (Cont.)

Part: Flanged Bronze Bearing Bushing | ***Validates concentric flanges, cylindrical body subtraction (ID bore), and circular pitch (PCD) bolt holes.***

STEP 1 Creation	`Create a flanged bushing with an outer body diameter of 45mm, length 60mm, a flange diameter of 70mm, flange thickness 12mm, and a 30mm center bore`
STEP 2 Edit	`Add 4 mounting holes of 6mm diameter evenly spaced on a 55mm bolt circle diameter on the flange`
Expected	Sleeve bearing with top retaining flange and 4 polar-array mounting holes.

3. Piping, Hydraulics & Fluid Systems

Part: High-Pressure 8-Bolt Pipe Flange | ***Validates ANSI/DIN circular bolt circle distribution, central fluid conduit, and planar face cuts.***

STEP 1 Creation	`Create an ANSI-style pipe flange with outer diameter 150mm, thickness 18mm, and a 60mm center through-hole`
STEP 2 Edit	`Add 8 bolt holes of 12mm diameter evenly spaced on a 115mm bolt circle diameter`
Expected	Standard 8-bolt flange disk ready for pipe welding or gasket mating.

Part: Hydraulic Distribution Manifold Block | ***Validates blind counter-bores, linear hole arrays, top edge filleting, and multi-port fluid routing.***

STEP 1 Creation	`Create a solid manifold block 120mm long, 90mm wide, and 50mm high with 6mm fillets on top edges`
STEP 2 Edit	`Add three 18mm diameter ports spaced 30mm apart on the top face to a depth of 35mm, and 4 corner mounting holes of 8mm through the block`
Expected	Hydraulic valve subplate with 3 blind fluid cavity ports and 4 through mounting bolt locations.

4. Couplings & Motion Enclosures

Part: Rigid Split-Hub Clamp Coupling | *Validates thin slit cutting, perpendicular cross-axis bore drilling, and clamping mechanics.*

STEP 1 Creation	`Create a cylindrical coupling sleeve with 50mm outer diameter, 80mm length, and a 20mm center shaft bore`
STEP 2 Edit	`Cut a 2mm wide lengthwise slit along one side of the cylinder and add two 6mm pinch bolt holes perpendicular to the bore`
Expected	Cylindrical shaft coupler with tensioning relief slot and transverse clamping screw holes.

Part: Aluminum Electronic Device Enclosure Base | *Validates pocket shelling, corner standoffs with tapped pilot holes, and exterior wall chamfers.*

STEP 1 Creation	`Create an enclosure box 140mm long, 90mm wide, and 35mm high with a wall thickness pocket hollowed out to 30mm depth`
STEP 2 Edit	`Add 4 internal corner standoff bosses 10mm diameter by 25mm high with 3mm center screw holes`
Expected	Machined/cast electronics chassis with interior PCB mounting standoffs.

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5. Step-by-Step Live Testing Procedure

Action 1: Launch Workbench	Open Google Chrome and navigate to: **http://192.168.11.86:9999**
Action 2: Execute Step 1	Copy any **STEP 1** prompt from above, paste it into the prompt bar at the bottom of the screen, and press Enter.
Action 3: Verify Browser	Within **~0.30 seconds**, the solid model will generate via the fast-path B-Rep engine and render in the interactive Three.js 3D viewport.
Action 4: Check Inventor	Watch the Windows Workstation monitor (192.168.11.150). Within 2–3 seconds, **Autodesk Inventor 2025 automatically pops to the foreground** with the newly generated .ipt model open and saved in C:\Users\koustubh\Documents\OmniCAD\.
Action 5: Conversational Edit	In the *same project conversation* on the web UI, paste the corresponding **STEP 2** command and hit Enter. Observe that the existing solid updates seamlessly without topology loss.